Ezau Faridh Torres Torres

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Applied mathematician (MSc @ CIMAT) specialized in NLP, deep learning, and data science. Experienced in real-world text analytics and scientific computing, with a strong foundation in math and ML.

Experience

Research assistant — *Guanajuato, Gto.*
Center for Research in Mathematics, A.C. (CIMAT) — *Aug 2024 – Jul 2025*

- Contributed to a research project on Bayesian Uncertainty Quantification developing surrogate models with PINNs in PyTorch and employed probabilistic models for uncertainty analysis.
- Co-authored a paper in preparation on deep Bayesian surrogate modeling applied to scientific simulations.

Research assistant — *San Luis Potosí, SLP.*
Potosino Institute of Scientific and Technological Research, A.C. (IPICYT) — *Mar 2022 – Jan 2023*

- Contributed to a research project analyzing mathematical models of neuronal responses to external stimuli. Conducted simulations and sampling of probability distributions modeling neuron action rates and applied advanced mathematical techniques to improve the understanding of neuronal dynamics.

Education

Center for Research in Mathematics, A.C. (CIMAT) — *Aug 2023 - June 2025*
MSc in Applied Mathematics. GPA: 9.6/10.0
Thesis: Physics-Informed Neural Networks for Bayesian Inverse Problems (Expected: July 2025)

Autonomous University of San Luis Potosí (UASLP) — *Aug 2017 – Dec 2021*
BSc in Applied Mathematics. GPA: 9.1/10.0

Technical Proficiencies

Programming Languages: Python, R, MATLAB, SQL, Git, Wolfram Mathematica, wxMaxima, C++.

Technologies & Tools: PyTorch, TensorFlow, Keras, Scikit-learn, Jupyter, Pandas, NumPy, Object-oriented programming, SciPy, Matplotlib, HuggingFace, transformers, fastText, GitHub, Google Cloud, DBeaver, Seaborn, LaTeX, Linux.

Languages: English IELTS C1.

Relevant Coursework

- **Deep Learning & Machine Learning:** Natural Language Processing, LLMs, IA & Advanced Deep Learning, Big Data, Pattern Recognition, Machine Learning techniques and applications, Optimization, Scientific Computing for Data Science, Computer Vision, Data Preparation, Data Manipulation, Model Optimization, Predictive Modeling, Algorithms.
- **Mathematics & Modeling:** Mathematical Modeling, Advanced Numerical Methods, Numerical Analysis, Dynamical Modeling, Analytical Modeling, Analytical Stochastic Processes, Matrix Methods for Data Analysis.
- **Statistics & Analysis:** Probabilistic and Statistical Models, Statistical Analysis, Bayesian Inference, Uncertainty Quantification (MSc Thesis topic), Markov Chain Monte Carlo Methods.
- **Other Technical Areas:** Geometry and Computer-Aided Design, Optimal Control in Robotics (BSc Thesis topic).

Portfolio Projects

Rest-Mex 2025 – NLP Challenge — *Github*

- Multitask classification of Spanish-language tourist reviews using Hierarchical Attention Networks (HAN).
- Ranked in the top 10% using Hierarchical Attention Networks (HAN) to classify sentiment, destination type, and Magic Town from Spanish-language reviews.

PINNs Framework (in progress) — *Github*

- Modular solver for PDEs with Physics-Informed Neural Networks.
- Built a PyTorch-based framework to solve forward and inverse PDEs in several domains, incorporating uncertainty quantification with MCMC.

Multitask Tweet Classification — *Github*

- Joint prediction of gender and nationality from Spanish tweets.
- Combined RoBERTuito embeddings and TF-IDF features in a hybrid model trained with multitask loss and evaluated via joint accuracy.

Relevant Courses & Certificates

- Google Advanced Data Analytics Professional Certificate — Google Career Certificates.
- Data Science School (UNAM) · Industrial Problem-Solving Workshop (CIMAT) · Numerical Modeling School (CIMAT)